

EEE/CPE 64 — Lab Report 1

BASIC LOGIC GATES

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INTRODUCTION

INTRODUCTION

Logic Gates are fundamental part of digital electronics. They are used in various things in embedded systems, computer, chips, communication systems, etc... Learning Logic gates are essential to understand how computer and various electronic system works.

In this section, the purpose of the lab is to define all the 7 basic logic gates in my words to show my understanding and use software (NI Multisim) to create a schematic of the circuit and simulate all the logic gates, provide images of the following simulation, and gather all the output data in a truth table of the logic gates. Finally, make a circuit on a breadboard to confirm all the output data and verify whether the simulation outputs are valid. Due to limited breadboard space, only three gates (OR, AND, NAND) will be implemented.

PRELIMINARY CALCULATIONS

2.1 DEFINITION OF SEVEN BASIC LOGIC GATES SYMBOLS

- a) **NOT**: Negates the input and outputs the negated value.
- b) **AND**: Outputs true only when both inputs are true.
- c) **NAND**: Negates the output value of the AND gate.
- d) **OR**: Outputs true if one or both of the inputs are true.
- e) **NOR**: Negates the output value of the OR gate.
- f) **XOR**: Outputs true only when the two inputs are different.
- g) **XNOR**: Negates the output of the XOR gate.

Logic Gates

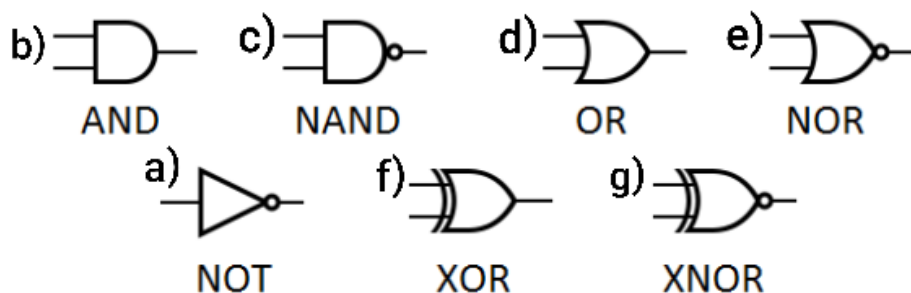


Figure 1: Shows all the 7 basic Logic Gates

SIMULATION RESULTS

3.1 NI MULTISIM: SCHEMATICS

In this part, I used *NI Multisim* software to place all seven basic logic gates individually using the “Place” menu. A DSWPK_10 component (labeled S1) was added to control two individual inputs, and a BAR_LED_BLUE_TEN block (labeled LED2) was used to observe the outputs of the logic gates. A digital power source (VDD) supplied 5 V to the circuit, resistors were included to protect the LEDs, and a common ground was connected to complete the circuit, as shown in Figure 2.

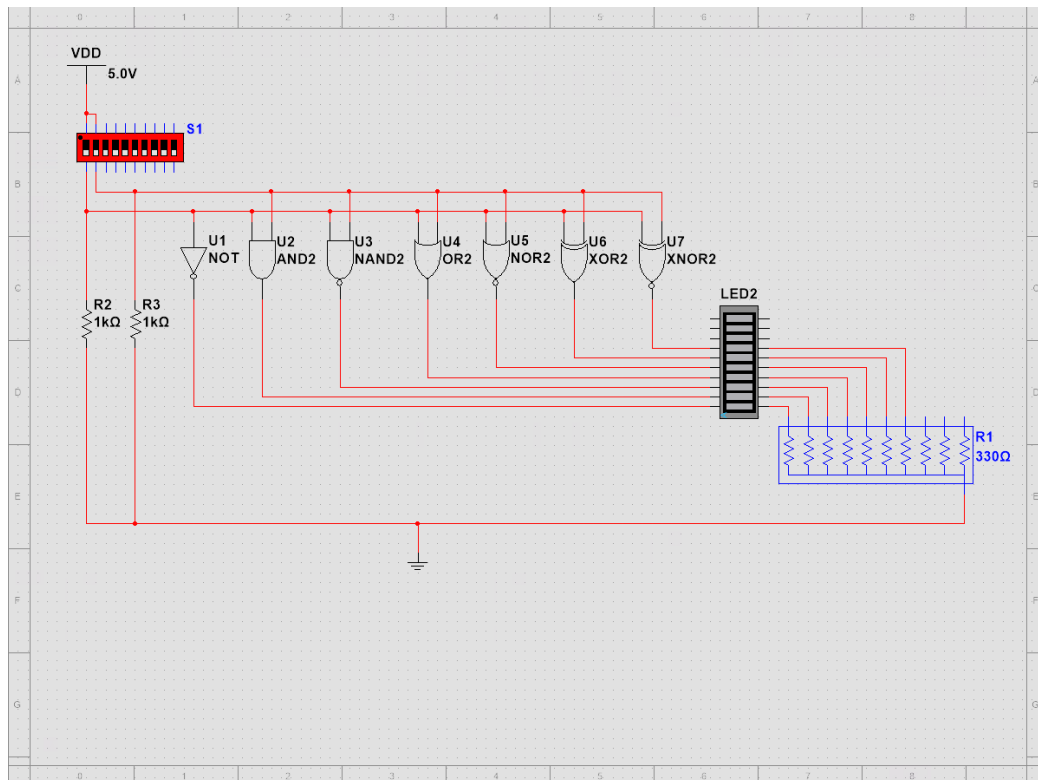


Figure 2: Schematic showing the seven logic gates and related components.

Components used

Supply: VDD (5 V digital source).

Inputs: DSWPK_10 (labeled S1).

Gates: NOT, AND, NAND, NOR, OR, XOR, XNOR.

Indicators: BAR_LED_BLUE_TEN (labeled LED2) with series resistors.

Resistor: Resistor and Bus Resistor.

3.2 VERIFY THE TRUTH TABLE

Once the circuit was set up, the simulation was run and the output data were collected. In Figure 3, the blue LED indicates the output of the logic gate. The output is considered 1 (true) when the LED is on and 0 (false) when it is off. For clarity, the first switch (S1_1) is defined as input A, and the second switch (S1_2) is defined as input B. As an example, when both switches are off (A=0, B=0), the corresponding output can be observed as shown below. Here is the first data that I acquired when both switches are 0 (Off).

A	B	NOT A	AND	NAND	OR	NOR	XOR	XNOR
0	0	1	0	1	0	1	0	1

Table 1: The Result of The First Truth Table.

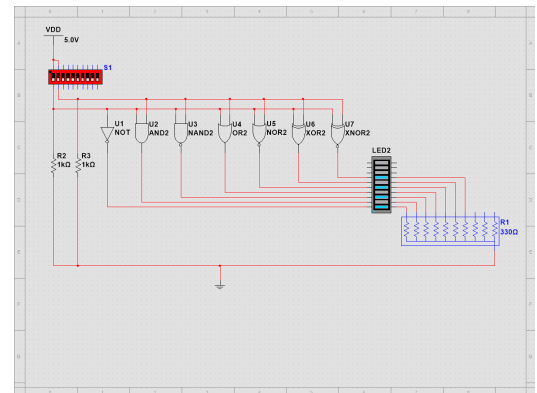


Figure 3: Schematic showing the seven logic gates and related components.

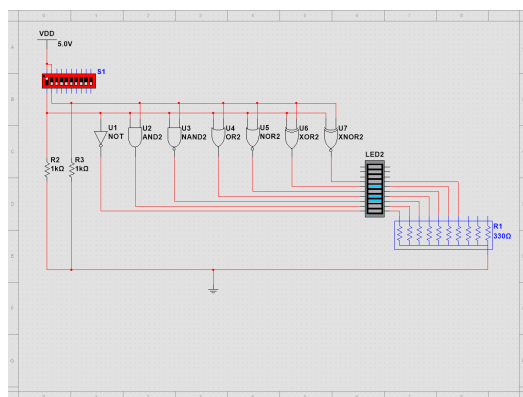


Figure 4: Schematic showing the seven logic gates and related components.

Now that we have seen the first Truth Table 1, we can see accordingly that when Switch A and B is 0 (off) some of the output values of the logic gates have been negating and resulting in opposite value of the corresponding input as shown in Figure 3. Let us see the result on Truth Table 2 after the switch A is 1 (on) and the switch is 0 (off).

A	B	NOT A	AND	NAND	OR	NOR	XOR	XNOR
1	0	0	0	1	1	0	1	0

Table 2: The Result Of The Second Truth Table

With the next input configuration, where switch (A=0) (off) and switch (B=1) (on), the circuit produces the outputs summarized in Table 4. The schematic used to obtain these results is shown in Figure 5.

A	B	NOT A	AND	NAND	OR	NOR	XOR	XNOR
0	1	1	0	1	1	0	1	0

Table 3: The Result of The Third Truth Table.

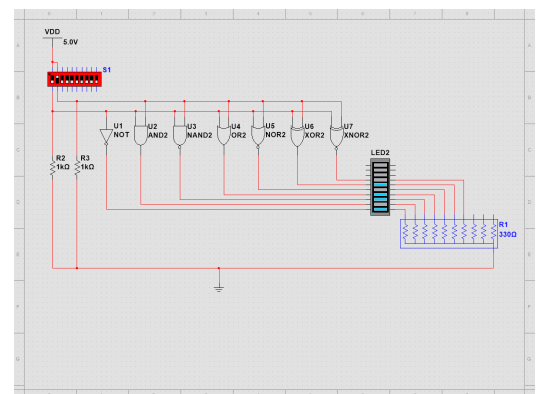


Figure 5: Schematic showing the seven logic gates and related components.

With the next input configuration, where switch $A = 0$ (off) and switch $B = 1$ (on), the circuit produces the outputs summarized in Table 4. The schematic used to obtain these results is shown in Figure 5.

A	B	NOT A	AND	NAND	OR	NOR	XOR	XNOR
1	1	0	1	0	1	0	0	1

Table 4: The Result of The Third Truth Table.

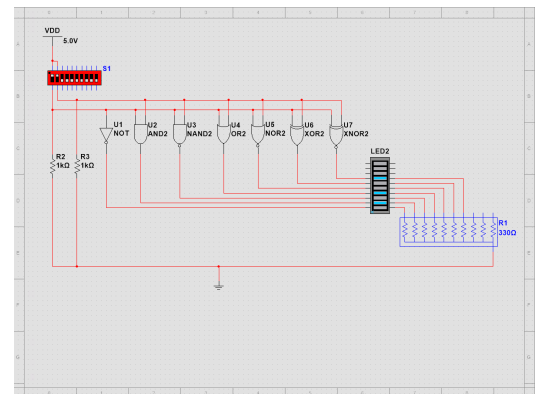


Figure 6: Schematic showing the seven logic gates and related components.

3.3 SIMULATION DATA

The simulation data collected from NI Multisim is summarized in Table 5. For each of the four possible input combinations ($A=0, B=0$; $A=1, B=0$; $A=0, B=1$; $A=1, B=1$), the outputs of all seven basic logic gates were observed and recorded. The results align exactly with the theoretical definitions introduced in the preliminary section. For example, the AND gate only produces a 1 when both inputs are high, while the NAND gate shows the opposite output as expected. Similarly, the XOR gate outputs 1 only when the inputs differ, and the XNOR gate gives the complementary value.

A	B	NOT A	AND	NAND	OR	NOR	XOR	XNOR
0	0	1	0	1	0	1	0	1
1	0	0	0	1	1	0	1	0
0	1	1	0	1	1	0	1	0
1	1	0	1	0	1	0	0	1

Table 5: Truth Table Result Data.

BREADBOARD RESULTS

4.1 BREADBOARD DATA

In this section, the simulation results are verified by implementing the circuits on a physical breadboard. Due to limited space, only three of the seven logic gates (OR, AND, and NAND) were constructed. The setup involved placing the components, input switches, resistors, and Bar LEDs on the breadboard, powered by a regulated 5V supply. The LEDs indicate the gate outputs in the same way as the NI Multisim simulations: 1 (ON) when the LED lights up and 0 (OFF) when it remains dark.

Figures 7–10 show the results of the breadboard implementation. The first gate tested is the OR gate, the second is the AND gate, and the third is the NAND gate. The fourth image shows the complete breadboard wiring with all three gates connected. This illustrates how the individual circuits can be integrated onto a single breadboard while maintaining correct functionality.

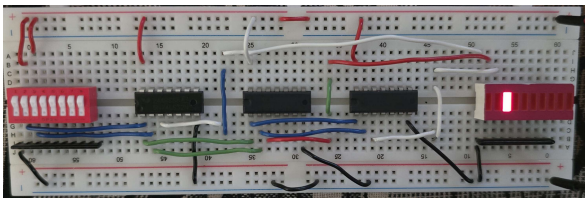


Figure 7: Caption for Image 1

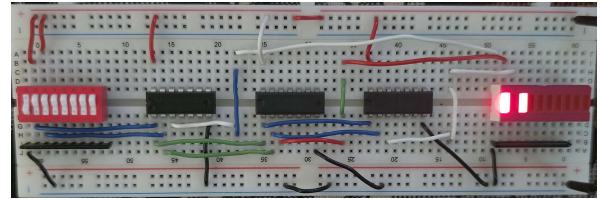


Figure 8: Caption for Image 2

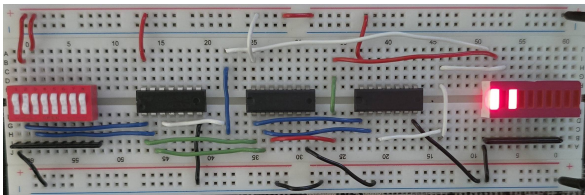


Figure 9: Caption for Image 3

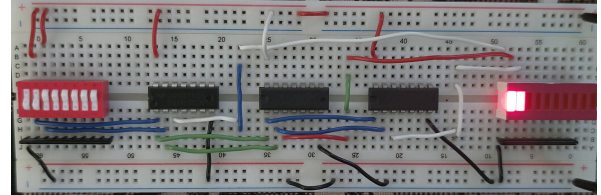


Figure 10: Caption for Image 4

A	B	OR	AND	NAND
0	0	0	0	1
0	1	1	0	1
1	0	1	0	1
1	1	1	1	0

Table 6: Observed breadboard truth table for OR, AND, and NAND gates.

4.2 CONCLUSION

Conclusion, The breadboard implementation of the OR, AND, and NAND gates was successful. The measured outputs matched exactly with the expected theoretical values and the Multisim simulations, confirming the accuracy of both the design and the experimental setup. The data collected demonstrated that the gates operated as defined: the OR gate produced a high output when at least one input was high, the AND gate produced a high output only when both inputs were high, and the NAND gate provided the inverted AND result. Overall, the experiment verified the correctness of digital logic principles in both simulation and hardware.